CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT

Course Code and Title**: ITCS214LA – DATA STRUCTURES AND ALGORITHM [LABORATORY]**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

DIRECTIONS: Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, as you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

Format, parts, and description	Compliance	
	Complied With	Not Complied With
Format		
1. The 2023-2024 syllabus format is used.	/	
2. The font used in the syllabus preparation is Times New Roman (11).	/	
3. DLSUD (being first) and college logos are beside each other at the top (center).	/	
4. Preliminaries include all the contact details of the professor, such as the day and time.	/	
5. Each WEEK has a corresponding date.	/	
Content		
6. The Learning Outcomes have more HOTS than LOTS.	/	
7. CLOs and TLOs are identified for each module/lesson/topic.	/	
8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities.	/	
9. The journal articles/references are within the last five years.	/	
10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies.	/	
11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students.	/	
Assessments		
12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus.	/	
13. Formative Assessments are not reflected anymore in the syllabus.	/	
14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities.	/	
15. The syllabus presents one-module(lesson)-one-assessment correspondence.	/	
16. Only one Enabling Assessment is reflected in a given week	/	
17. The Enabling Assessments are aligned with the identified CLOs and TLOs.	/	
18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term.	/	
19. Thirty percent (30%) is allotted for Summative Assessments per term.	/	

20. Twenty percent (20%) is allotted for Class Participation/Attendance per term.	/	
21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade.	/	

22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching- learning activities.	/	
23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching- learning activities.	/	
24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD	/	
25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD.	/	
References and Curriculum Map		
26. The References must be within the last 5 years except for canons and seminal studies.	/	
27. It must be a combination of books, eBooks, journal articles, videos, etc.	/	
28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS.	/	
29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs)	/	

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

Self-reviewed:

Orlando A. Zapanta Faculty

Checked, verified, and endorsed:

Rolando B. Barrameda
Dept. Curriculum Committee Reviewer

Checked, verified, and approved:

Josephine T. Eduardo
Department Chair

Dr. Rubie Causaren
College Dean

DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF SCIENCE AND COMPUTER STUDIES

COMPUTER STUDIES DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITCS214 (CS)
COURSE TITLE	**:**	Data Structures and Algorithms
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	2 Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	S-ITCS123 / S-ITCS123LA
CO-REQUISITES	**:**	S-ITCS123LA
PROFESSOR	**:**	Orlando A. Zapanta (oazapanta@dlsud.edu.ph)
CONSULTATION HOURS	**:**	Monday / 9:00-11:00

COURSE DESCRIPTION:

This course provides the concepts of abstract data types of common data structures such as lists, linked lists, stacks, queues, trees and graphs used in solving computer problems. This course also includes the study of time and space complexities of algorithms and the choice of appropriate data structures. The course covers several data structures as tools for programming, best practices in using those data structures with estimated running time and space efficiency, hands-on exercises to develop practical skills in creating impactful programs

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:

CLO1. Understand the importance of data structures and algorithms in application of daily living.
CLO2. Evaluate the running time of algorithms and data structures using big-oh notations.
CLO3. Analyze the features of each data structure and its operations and use it to design algorithms to solve real-world problems.
CLO4. Apply the most appropriate data structures given a specified real world problem.
CLO5. Create codes written in Java for various algorithms and data structures.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN FOR MIDTERM PERIOD:

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand data structures & algorithms in application of daily living.	**TLO1.** Distinguish the differences between algorithms and data structures. **TLO2.** Identify the criteria for efficient algorithms.
CLO2. Evaluate the running time of algorithms and data structures using big-oh notations.	**TLO3.** Calculate the running time of codes following specific rules.
CLO3. Analyze the features of each data structure and its operations and use it to design algorithms to solve real-world problems.	**TLO4.** Construct pseudo-language implementations of lists and its operations. **TLO5.** Write algorithms for linked list and its variations. **TLO6.** Understand the concept of a stacks and its operations.
CLO6. Demonstrate 21st century skills in all learning activities.	**TLO7.** Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem- Solving, Use of ICT (not part of topic, but integrated in assessments)

LEARNING OUTCOMES FOR MIDTERM PERIOD

Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy Module Activities Online/ Offline	Technology Enabler	Asynchronous	Onsite	Allotted Hours
Module 0 CLO1 TLO1	**W1** Aug 14 – 19	Gospel Reading Presentation of the course syllabus **Module 0 – Course Introduction** Introduction of the course professor Presentation of the Enrich Virtual Learning model LMS Formative Activity: Data Structures Primer https://www.youtube.com/watch?v=bum_19loj9A&list=PLBZBJbE_rGRV8D7XZ08LK6z-4zPoWzu5H Class Participation Activity	Classroom PowerPoint YouTube		✓	2
Module 1 CLO1 TLO2	**W2** Aug 22 – 26	Gospel Reading **Module 1 – Introduction to Data Structures** Module 1 Enabling Assessment about Data Structures	Schoolbook PowerPoint Quizziz		✓	2
Module 2 TLO3, CLO2, CLO6	**W3** Aug 29 – Sep 2	Gospel Reading **Module 2 – Introduction to Algorithm Analysis** Module 2 Enabling Assessment about Algorithm Analysis	Schoolbook PowerPoint Canva	✓		2

Module 3 CLO2 TLO3	**W4** Sep 4 -9	Gospel Reading **Module 3 – List as an abstract data type** Module 3 Enabling Assessment about Data Types	Schoolbook PowerPoint Wooclap	✓		2
Module 4 TLO5, CLO3, CLO5, CLO6	**W5** Sep 11 – 16	Gospel Reading **Module 4 – Linked List as an abstract data type** Module 4 Assessment about Lists and Array	Schoolbook Canva		✓	2
Module 4 TLO5, CLO3, CLO5, CLO6	**W6** Sep 18 – 23	Gospel Reading **Module 4 – Linked List Applications** Module 4 Enabling Assessment about Linked List	Schoolbook Canva	✓		2
Module 5 TLO6, CLO3, CLO5, CLO6	**W7** Sep 24 – 29	Gospel Reading **Module 5: Stack as an abstract data type** Module 5 Assessment about Stacks	Schoolbook PowerPoint Wakelet Quizlet		✓	2
Module 5 TLO6, CLO3, CLO5, CLO6	**W8** Oct 2 – 7	Gospel Reading **Module 5: Stack Applications – Prefix, Infix, Postfix Notations; Postfix Evaluation** Module 5 Enabling Assessment about Stacks Application	Schoolbook PowerPoint Canva	✓		2
	W9 Oct 9 – 14	**MIDTERM SUMMATIVE ASSESSMENT** Coverage: Modules 1 to 5	Schoolbook		✓	2
			TOTAL	6	12	18

LEARNING PLAN FOR FINAL PERIOD:

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Apply the most appropriate data structures given a specified real world problem.	**TLO8.** Apply various applications of queue operations to real life scenarios. **TLO9.** Identify the different terminologies used in trees. **TLO10.** Explore different applications of binary trees. **TLO11.** Discover the varied ways of graph representations and applications. **TLO12.** Determine the steps in creating recursive routines.
CLO5. Create codes written in Java for various algorithms and data structures.	**TLO13.** Use and apply Java structure and elements to implement the algorithms and data structures covered in this course.
CLO6. Demonstrate 21st century skills inall learning activities.	**TLO7.** Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

LEARNING OUTCOMES FOR FINAL PERIOD

Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy	Technology Enabler	Asynchronous	Onsite	tted Hours
	W10 Oct 16 – 21	**STUDENTS' / COMPLETION WEEK** *Check the events posted for this purpose*	DLSU-D Schoolbook Newsfeed	✓		
	W11 Oct 23 – 28	**SELF-CARE WEEK** *Check the self-care hub posted for this purpose*	DLSU-D Self-Care Hub			
Module 6 TLO8, CLO4,	**W12** Oct 30 – Nov 4	Gospel Reading **Module 6: Queue as an abstract data type** Module 6 Enabling Assessment about Queue	Schoolbook PowerPoint		✓	2
Module 7 TLO9, CLO4, CLO5,	**W13** Nov 6 – 11	Gospel Reading **Module 7: Tree as an abstract data type** Module 7 Enabling Assessment about Trees	Classroom PowerPoint Canva	✓		2
Module 8 TLO10, CLO4, CLO5, CLO6	**W14** Nov 13 – 18	Gospel reading **Module 8: Binary Tree as an abstract data type** Module 8 Assessment about Binary Trees	Schoolbook PowerPoint Quizziz		✓	2
Module 8 TLO10, CLO4, CLO5, CLO6	**W15** Nov 20 – 25	Gospel reading **Module 8: Trees and Binary Trees' Applications** Module 8 Enabling Assessment about Binary Trees Application	Schoolbook PowerPoint Canva	✓		2

Module 9 TLO11, CLO4, CLO5, CLO6	**W16** Nov 27 – Dec 2	Gospel Reading **Module 9: Graph as an abstract data type** Module 9 Assessment about Graphs	Schoolbook Quizlet		✓	2
Module 10 TLO12, CLO4, CLO5, CLO6	**W17** Dec 4 – 9	**COMPLETION / CONSULTATION WEEK** *Students will comply with lacking assessments and/or in-person/online consultation with the professor*	Schoolbook PowerPoint Canva	✓		2
	W18 Dec 11 – 16	**FINAL SUMMATIVE ASSESSMENT** Coverage: Modules 6 to 10	Schoolbook PowerPoint Canva		✓	2
		TOTAL		10	8	18

GRADING SYSTEM

Midterm

Enabling Assessments (EA)	-	50%
Class Participation (CP)	-	20%
Summative Assessments (SA)	-	30%
Total		100%

Finals

Enabling Assessments (EA)	-	50%
Class Participation (CP)	-	20%
Summative Assessments (SA)	-	30%
Total		100%

Overall Final Grade = [(Midterm grade) + (Final term grade)] / 2

Note: Class participation (CP) refers to your active engagement in class from day 1 to the last day of each term (midterm or final). CP includes, but not limited to, the following: actual attendance, observance of netiquettes (synchronous) and etiquettes, promptness in submitting tasks/assessments, relevant questions asked, participation in collaborative tasks, etc.

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty

percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.

b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
    i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
    ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
    iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 15% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Call number or e-provider	Reference Material
QA 76.9.S88 .H191 2018	Mehta, D. P., & Sahni, S. (2018). Handbook of data structures and applications (Second edition.). CRC Press/Taylor & Francis Group.
QA 76.64 .D152 2018	Dale, N., Joyce, D. T., & Weems, C. (2018). Object-oriented data structures using Java (Fourth edition.). Jones & Bartlett Learning.

ONLINE REFERENCES

Call number or e-provider	Reference Material
e-Book	John Bullinaria 2019. Data Structures and Algorithms - School of Computer Science. Universityof Birmingham, UK. [accessed 2020 August 10] https://www.cs.bham.ac.uk.
e-Book	Goodrich, M. T. Data Structures and Algorithms in Java. Department of Computer Science University of California, Irvine. John Wiley & Sons, Inc. Retrieved from https://enos.itcollege.ee/~jpoial/algorithms/GT
YouTube Channel	Computer Education or all. 2018. Data Structures and Algorithms Complete Tutorial Computer Education for All [accessed 2022 August 15] https://www.youtube.com/watch?v=YWnBbNj_G-U
YouTube Channel	Tech Course PH. 2020. Data Structures and Algorithms Lecture Series. [accessed 2022 August 15] https://www.youtube.com/watch?v=CNhj4GZ_7iE

Prepared: **Computer Science Department**
1st Semester 2023 - 2024

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COLLEGE OF SCIENCE AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS INFORMATION TECHNOLOGY

COURSE CODE	**:**	S-ITCS214
COURSE TITLE	**:**	Data Structures and Algorithms
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	2 units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	ITCS123 / ITCS123LA
CO-REQUISITES	**:**	ITCS214LA

Course Learning Outcome	**Program Learning Outcomes**								
	PLO 1	**PLO 2**	**PLO 3**	**PLO 4**	**PLO 5**	**PLO 6**	**PLO 7**	**PLO 8**	**PLO 9**
1. Understand the importance of data structures and algorithms in application of daily living.	/	/	/	/	/				
2. Evaluate the running time of algorithms and data structures using big-oh notations.	/	/	/	/	/				
3. Analyze the features of each data structure and its operations and use it to design algorithms to solve real-world problems.	/	/	/	/	/				
4. Apply the most appropriate data structures given a specified real world problem.	/	/	/	/	/	/			
5. Create codes written in Java for various algorithms and data structures.	/	/	/	/	/	/	/	/	/

*NOTE: Provide **a check mark** on the areas in which the program learning outcome (PLO) is hit by the course*

learning outcome (CLO)

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GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME MAPPING TABLE FOR BS INFORMATION TECHNOLOGY

	1	2	3	4	5
1. Apply concepts of computing in different domains of information technology.				/	/
2. Identify and analyze the problem and define the computing requirements appropriate to its solution					/
3. Design, develop, test and evaluate various solutions using **modern computing tools** for complex computing problems that would take full advantage of usability of computer resources.			/	/	/
4. Document the computer-based systems by writing effective reports for end-user.				/	/
5. Work effectively as an individual or as a member of a team during the software development for company-based systems.			/	/	/
6. Manage and maintain system security, reliability and efficiency.				/	/
7. Communicate effectively with the computing community and non-**IT** professionals about computing activities	/	/	/		
8. Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and address these issues.					
9. Engage in continuing professional development through performing independent or collaborative research as computing professionals.			/	/	/

NOTE: Provide a check mark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expect from its graduates.

GRADUATES ATTRIBUTES AND INSTITUTIONAL LEARNING OUTCOMES (ILOs)

Graduate Attribute (GA)	Descriptors (Institutional Learning Outcome)[1]
1. God-centered	a) Creates an environment where the experience of God is lived and shared b) Practices honesty, fairness, truth and integrity in all aspects of life (personal and professional lives) c) Observes and maintains ethical standards in dealing with the different stakeholders d) Integrates Christian perspectives and values in all undertakings e) Manifests humility and respect in relating with other people
2. Patriotic	a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts c) Engages actively in political, social, economic, and cultural transformation for nation building d) Brings pride and honor to the community and the country e) Patronizes locally produced products and promotes them globally
3. Socially Responsive	a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation b) Engages in ecological advocacies as responsible stewards of God's creation c) Continuously works in solidarity with people and institutions to effect liberating action d) Brings forth awareness and deeper understanding of social realities e) Manifests spirit of volunteerism through social actions that improve communities
4. Globally Competitive	a) Takes progressive responsibility for own learning and development b) Promotes critical and creative thinking, self-knowledge and self-mastery c) Is committed to rise above mediocrity d) Shows dedication, commitment and love for work to attain exceptional results Exhibits innovativeness and creativity in various contexts

[1] *Approved in July 2017 during a multi-sectoral committee specifically convened for the purpose of coming up with descriptions for the graduate attributes.*